Cleaning and Merging Datasets

Methods

Human genetic sampling was conducted under informed consent procedure approved by the Pennsylvania State University and the Madagascar Ministry of Health. Participants were recruited from five localities in Madagascar, and each participant was asked their ethnic group and that of their grandparents. Saliva containing DNA was collected from each participant using Oragene collection tubes. DNA was extracted from saliva samples, and DNA concentrations were standardised in preparation for genotyping using Nanodrop.

Genetic and phenotypic data were collected by the team of Dr George Perry at Pennsylvania State University with the University of Antananarivo. Dr Jason Hodgson and Dr Rindra Rakotoarivony led this data collection in 2015. These data comprise of 192 individuals from five regions: Merina people from the Central Highlands (n = 36), Betsileo people from the southern Highlands (n = 38), Sakalava and Tsimihety people from the northwestern coast (n = 40), Betsimisaraka people from the eastern coast (n = 38), and people of various ethnic groups from the extreme north in the town of Diego (n = 40). ~690,000 SNPs were typed using the Illumina HumanOmniExpress-24 array.

Two main datasets were created, a Malagasy only dataset and a global comparative dataset. The Malagasy dataset was created by merging our data with Malagasy data from Pierron et al. (2014) (Mikea; n = 21, Temoro; n = 24, Vezo; n = 24). For a comparative global dataset, this data was combined with data from southeast Asian and African populations in the Auton et al. (2015) dataset (n = 882), Bantu African individuals from Fortes-Lima et al. (2024) (n = 1763), individuals from Borneo from Brucato et al. (2016) (n = 39), and individuals from the islands of Sumba (n = 30, subsampled at random from the full dataset of 240 individuals) and Timor (n = 7) Cox et al. (2016). The genome build of each dataset was verified using the function `snp_modifyBuild` from `bigsnpr` Privé et al. (2018) prior to merging to ensure each dataset used the same build (in this case, GRC37).

Data were merged in R using tidypopgen function `rbind` Carter and Manica (2025), which merges SNP datasets, resolves strand inconsistencies, and removes ambiguous SNPs. After merging, datasets were cleaned using tidypopgen Carter and Manica (2025). Only autosomal SNPs were analysed. All SNPs with a missingness rate above 0.05 across all samples were removed. All individuals with a genotype call rate less than 0.9 (missingness exceeding 0.1) were removed. SNPs with minor allele frequency below a threshold of 0.025 were removed. Hardy-Weinberg equilibrium p-values were calculated for each SNP within each population using the tidypopgen function qc_report_loci, which provides a Bonferroni correction given the number of populations analysed. SNPs with a p-value below 0.01 in at least one population after correction were removed. Close relatives were removed using the KING robust kinship coefficient Manichaikul et al. (2010), implemented in tidypopgen. One individual was removed from each pair with a kinship coefficient greater than 0.0884, equivalent to removing relatives up to the second degree. After this filtering, the Malagasy dataset consisted of 233 individuals and 596146 SNPs, and the global dataset consisted of 2744 individuals and 425931 SNPs.

SNPs in linkage were then removed by LD clumping Privé et al. (2018), using tidypopgen. Those with an r2 value greater than 0.2 in a window of 100 kilobases were removed. After LD clumping, the Malagasy dataset consisted of 227410 SNPs and the global dataset consisted of 198196 SNPs.

Principal components analysis (PCA) was used firstly to visualise the Malagasy dataset, and secondly the Malagasy individuals compared to global comparative data. Non-malagasy populations were randomly subsampled to a maximum of 30 individuals to avoid bias through uneven sample sizes. This left a total of 1955 individuals.

Unsupervised clustering using the model-based method fastmixture Santander, Refoyo Martinez, and Meisner (2024) was run on the full, cleaned dataset of 2744 individuals and 425931 SNPs using the package tidygenclust Tysall et al. (2025). Fastmixture has been shown to be computationally efficient for large datasets while providing comparable results to other model-based clustering algorithms such as STRUCTURE Pritchard, Stephens, and Donnelly (2000) and ADMIXTURE Alexander, Novembre, and Lange (2009), with reported speeds up to 30 times faster than ADMIXTURE Fumagalli (2024).

The algorithm was run for 9 values of K (2 to 10) and 3 runs per K with 5-fold cross validation. Subsequently, we analysed the fastmixture output using the Clumppling algorithm of Liu, Kopelman, and Rosenberg (2024), also implemented in tidygenclust Tysall et al. (2025). Clumppling aligns replicated runs of unsupervised clustering across multiple values of K to identify modes across replicates. A single mode was found for each value of K across 3 replicates.

f_2 statistics were calculated in tidypopgen Carter and Manica (2025), f_3 and f_4 ratio statistics were then calculated in Admixtools2 from these f_2 blocks Maier and Patterson (2024). Admixture f_3 statistics were used in the configuration f_3(Malagasy; Asian, African) to test whether the Malagasy population could be modelled as an admixture between the given African and Asian references.

As the Merina exhibited high inter-individual variation in fastmixture ancestry profiles, indicative of some form of population structure, we ran the supervised mode of ADMIXTURE Alexander, Novembre, and Lange (2009) at K = 2 for all Malagasy populations. Individuals from Borneo and individuals from Mozambique were used as reference populations in this supervised mode, as these reference populations were shown to give the minimum admixture f3 statistics values. Every Malagasy population showed inter-individual variation in ancestry proportions. However, supervised ADMIXTURE profiles revealed a bimodal distribution among the Merina Alexander, Novembre, and Lange (2009), which has been observed previously in this population Hodgson (2016). Merina with a high proportion of African ancestry and the Merina with a high proportion of Asian ancestry were then separated into two groups before running MALDER and pairwise FST analyses. Each individual was assigned to group Merina A or group Merina B based on their ancestry profile at k = 2 inferred using supervised ADMIXTURE Alexander, Novembre, and Lange (2009), with Merina A estimated to have >50% of ancestry derived from a population similar to Borneo individuals, and Merina B estimated to have >50% of ancestry derived from a population similar to Mozambique individuals. See supplementary Figure 2 for results of supervised ADMIXTURE analysis. Splitting this sample reduced power, as only 12 individuals remained in the higher African ancestry group. However, doing so avoided violating model assumptions (e.g the model of linkage employed by ALDER assumes that there is no structure within populations, that individuals are mating at random) and enabled us to consider whether there may be differing demographic histories among the Merina.

f_4 ratio statistics were used to examine the proportion of ancestry derived from Asian versus African sources for each population. To obtain these ratios, the global dataset was merged with 1000 Genomes GBR and a chimpanzee reference (PanTro3.0, GenBank accession GCA_000001515.5) to act as outgroups. f_4 ratios were calculated in the configurations f_4(GBR,Chimpanzee;Malagasy,Borneo) / f_4(GBR,Chimpanzee;Esan,Borneo) and f_4(GBR,Chimpanzee;Malagasy,Esan) / f_4(GBR,Chimpanzee;Borneo,Esan).

To obtain a chimpanzee reference, first SNPs in our dataset were lifted to PanTro3.0 (GenBank accession GCA_000001515.5) from hg19 using liftOver overchain file https://hgdownload.soe.ucsc.edu/goldenPath/hg19/liftOver/hg19ToPanTro5.over.chain.gz from UCSC Genome Browser Kent et al. (2002), through bigsnpr function snp_modifyBuild() Privé et al. (2018). The chimpanzee reference for each SNP was then obtained using a custom R function to query the Ensembl API (GenBank accession GCA_000001515.5, Ensembl release 115, accessed via Ensembl API https://rest.ensembl.org/documentation/info/sequence_region on 17-12-2025). This procedure retained 77993 SNPs for f_4 ratio calculation. Standard errors were calculated using block-jackknife resampling (admixtools f_4 ratio default). See Supplementary Table 3 for full f_4 ratio output.

MALDER Loh et al. (2013); Pickrell et al. (2014) was used to date the timing of admixture across different Malagasy groups. Genetic distances for each SNP were set using HapMap Phase II genetic map Frazer et al. (2007), mapped to build GRCh37. tidypopgen was used to create a subset of the full dataset containing only SNPs with known genetic map positions in the correct order. This dataset was not pruned for linkage

disequilibrium, and consisted of 236715 SNPs. MALDER was run using the default -binsize of 5 * 10-4 Morgans (describing the size of the bins that SNPs are split up into. Instead of directly calculating distance between each SNP pair, MALDER uses the distance between the bins of each SNP pair.), and the –mindis parameter (minimum genetic distance to begin LD decay curve fitting) set to 0.005 cM. Standard error of estimated dates were calculated through leave-one-chromosome-out jackknifing, with the flag -jackknife.

Pairwise F_{st} distances between Malagasy populations were calculated in tidypopgen, using the method of Weir and Cockerham (1984).

Supplementary Data

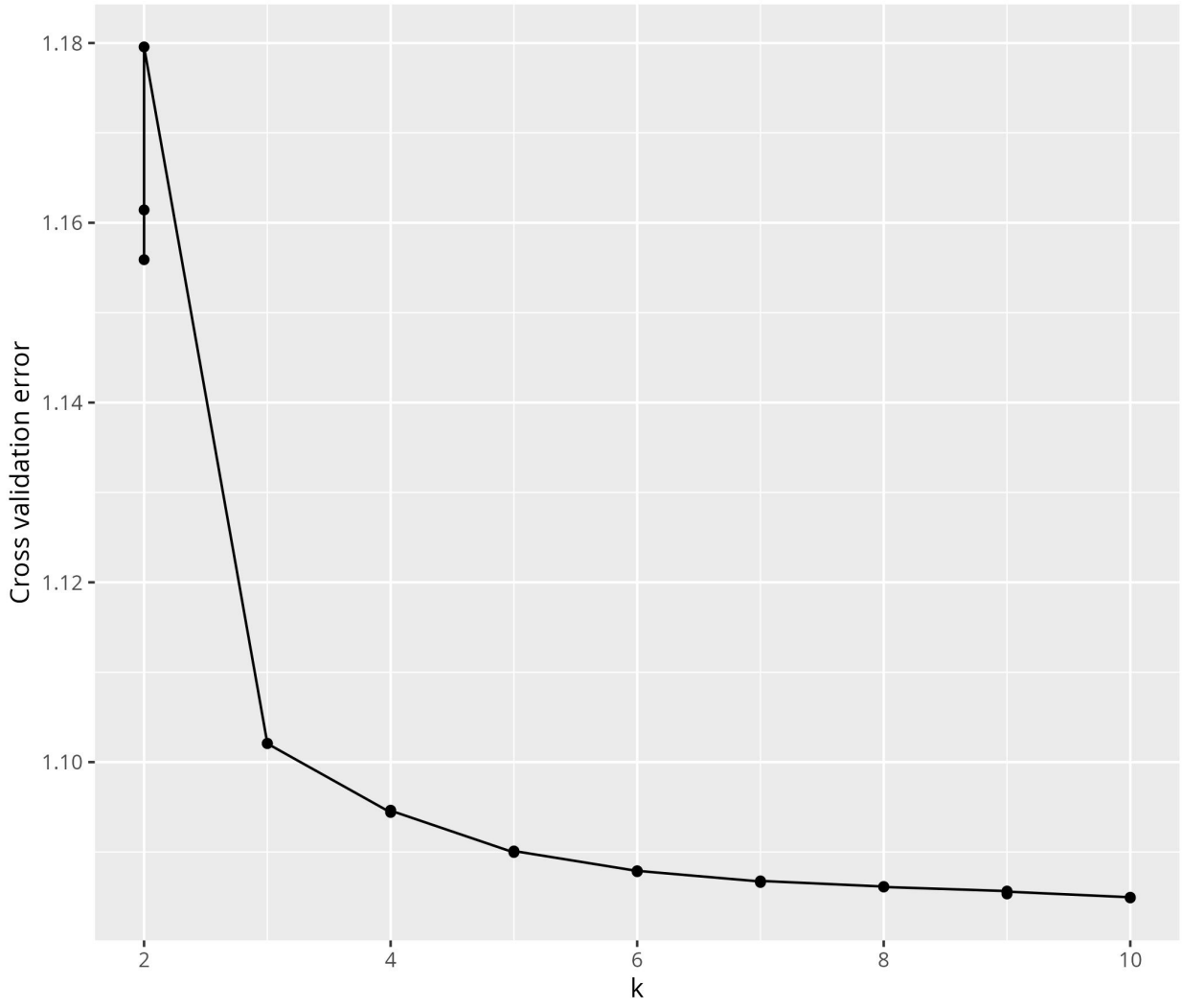


Figure 1: Supplementary Figure 1: Cross-validation error for fastmixture.

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## Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
## resized.
```

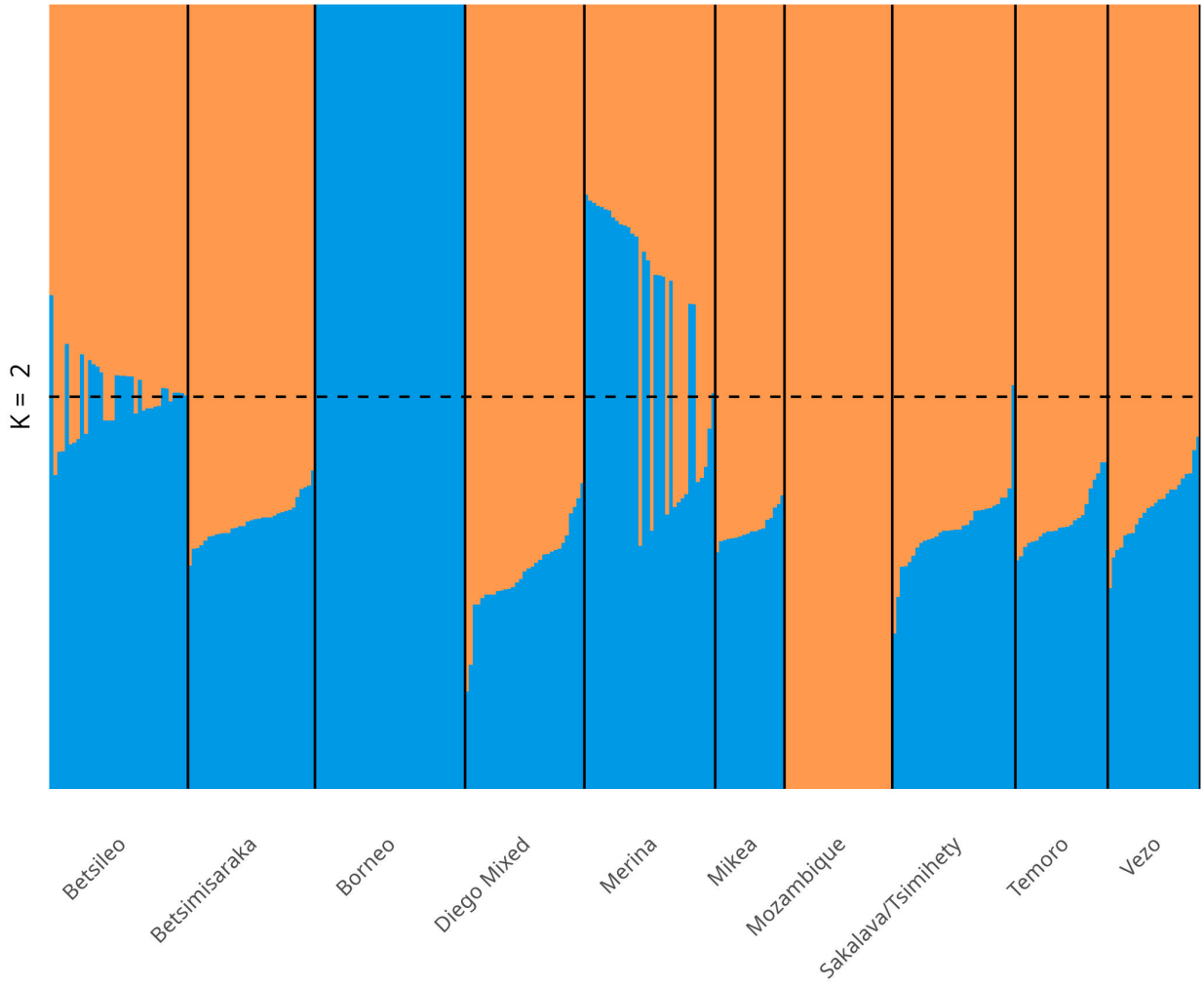


Figure 2: Supplementary Figure 2: Supervised admixture at K = 2 for for Malagasy compared to reference populations. Individuals from Borneo from Brucato et al. (2016), and individuals from Mozambique populations from Fortes-Lima et al., (2024). Mozambique individuals are labelled together, but this group contains individuals given the following population labels: “Mozambique_Nhungwe”, “Mozambique_Tsonga”, “Mozambique_Ronga”, “Mozambique_TongaMoz”, “Mozambique_Chuwabu”, “Mozambique_Tswa”, “Mozambique_Chopi”, “Mozambique_Sena”, “Mozambique_Makhuwa”.

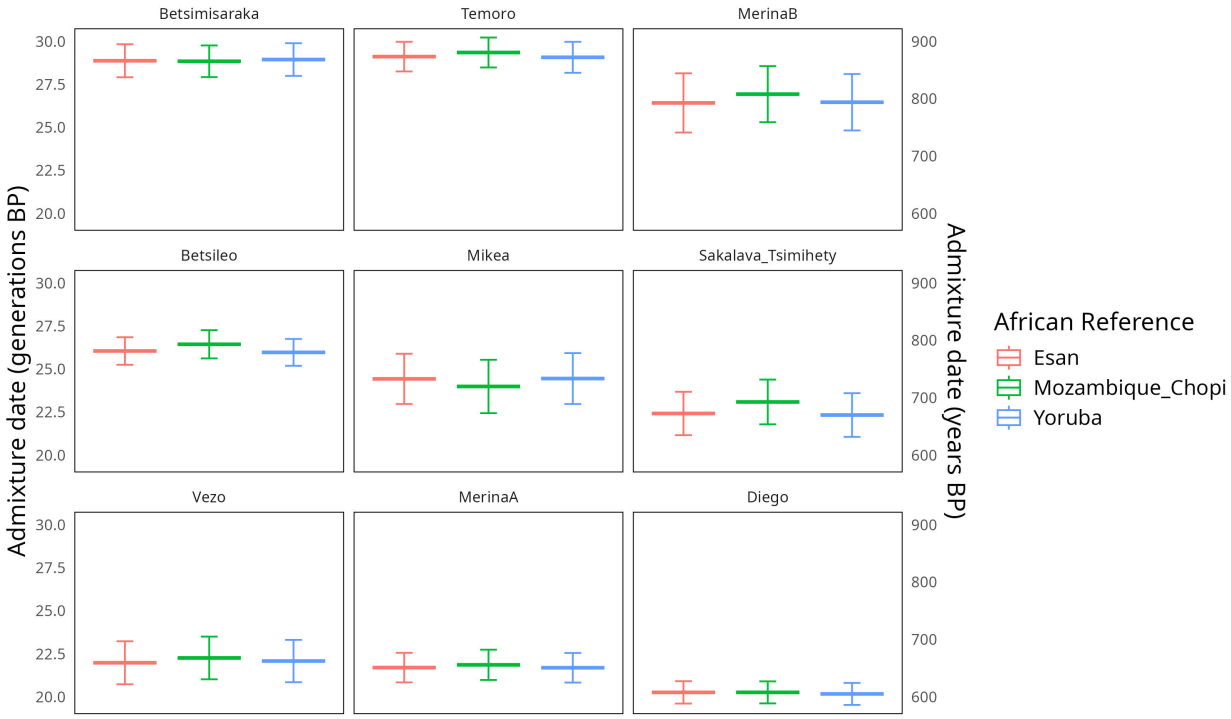


Figure 3: Supplementary Figure 3: Estimated admixture dates for Malagasy populations using MALDER (Pickrell et al., 2014), with Borneo as an Asian source population and varied African source populations. Horizontal bar indicates estimated date of admixture, vertical bars represent standard error from leave-one-chromosome-out jackknifing.

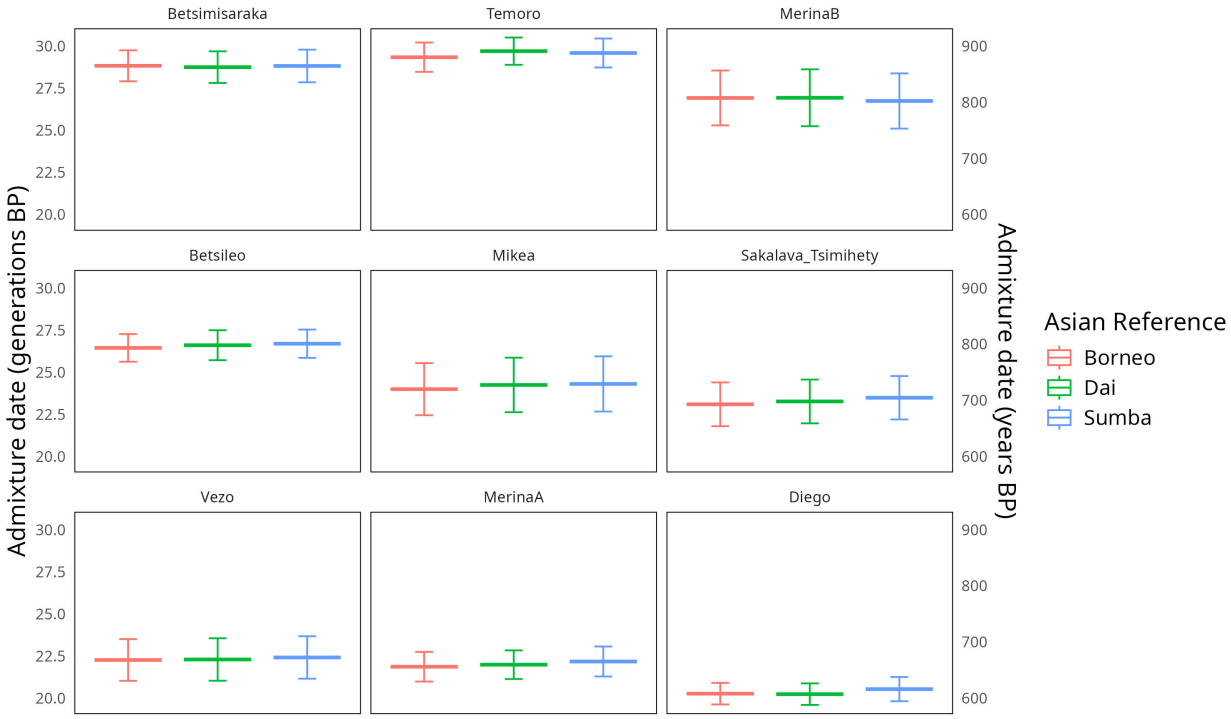


Figure 4: Supplementary Figure 4: Estimated admixture dates for Malagasy populations using MALDER (Pickrell et al., 2014), with Mozambique as an African source population and varied Asian source populations. Horizontal bar indicates estimated date of admixture, vertical bars represent standard error from leave-one-chromosome-out jackknifing.

Table 1: Supplementary Table 4: MALDER results for all Malagasy pop
lations, with varied African and Asian source populations. Columns are
follows: ...

test_res	p_value	test_pop	ref A	ref B	2-ref z-score	2-ref decay	2-ref amp_exp
success	3e-10	Betsileo	Borneo	Sumba	6.30	55.59 +/- 8.83	0.00002570 +/- 0.00000408
success	0.027	Betsileo	Borneo	Dai	2.21	212.86 +/- 64.59	0.00001889 +/- 0.00000854
success	4.6e-222	Betsileo	Borneo	Esan	31.81	26.05 +/- 0.80	0.00216033 +/- 0.00006791
success	3.6e-234	Betsileo	Borneo	Yoruba	32.67	25.97 +/- 0.78	0.00214609 +/- 0.00006568
success	5.7e-230	Betsileo	Borneo	Mozambique_Chopi	32.38	26.44 +/- 0.82	0.00227784 +/- 0.00006829
success	6e-06	Betsileo	Sumba	Dai	4.53	59.66 +/- 13.18	0.00003018 +/- 0.00000645
success	3.5e-188	Betsileo	Sumba	Esan	29.26	26.26 +/- 0.82	0.00191612 +/- 0.00006549
success	1.7e-196	Betsileo	Sumba	Yoruba	29.91	26.18 +/- 0.80	0.00190193 +/- 0.00006360
success	1.5e-200	Betsileo	Sumba	Mozambique_Chopi	30.22	26.69 +/- 0.84	0.00202856 +/- 0.00006714
success	8.6e-200	Betsileo	Dai	Esan	30.16	26.21 +/- 0.87	0.00216652 +/- 0.00006781
success	1.5e-205	Betsileo	Dai	Yoruba	30.59	26.14 +/- 0.85	0.00215206 +/- 0.00006551
success	1.3e-197	Betsileo	Dai	Mozambique_Chopi	29.99	26.60 +/- 0.89	0.00228532 +/- 0.00006899
failure	0.35	Betsileo	Esan	Yoruba	0.94	143.91 +/- 103.08	0.00000134 +/- 0.00000143
success	1.8e-05	Betsileo	Esan	Mozambique_Chopi	4.29	116.72 +/- 27.20	0.00003934 +/- 0.00000830
success	6.1e-06	Betsileo	Yoruba	Mozambique_Chopi	4.52	116.46 +/- 25.76	0.00003943 +/- 0.00000786
success	6.2e-15	Betsimisaraka	Borneo	Sumba	7.80	65.04 +/- 7.84	0.00002275 +/- 0.00000292
success	0.043	Betsimisaraka	Borneo	Dai	2.03	168.56 +/- 83.20	0.00001141 +/- 0.00000551
success	1.1e-196	Betsimisaraka	Borneo	Esan	29.92	28.87 +/- 0.96	0.00187297 +/- 0.00004607
success	7.2e-204	Betsimisaraka	Borneo	Yoruba	30.47	28.94 +/- 0.95	0.00186972 +/- 0.00004541
success	6.9e-217	Betsimisaraka	Borneo	Mozambique_Chopi	31.43	28.84 +/- 0.92	0.00197274 +/- 0.00004997
success	1.6e-07	Betsimisaraka	Sumba	Dai	5.24	70.95 +/- 12.86	0.00002673 +/- 0.00000510
success	4.5e-185	Betsimisaraka	Sumba	Esan	29.01	28.90 +/- 1.00	0.00165692 +/- 0.00005122
success	2.2e-189	Betsimisaraka	Sumba	Yoruba	29.35	28.96 +/- 0.99	0.00165296 +/- 0.00005075
success	1.6e-193	Betsimisaraka	Sumba	Mozambique_Chopi	29.68	28.83 +/- 0.97	0.00175133 +/- 0.00005520
success	1.3e-199	Betsimisaraka	Dai	Esan	30.14	28.81 +/- 0.96	0.00186998 +/- 0.00004787
success	6.6e-205	Betsimisaraka	Dai	Yoruba	30.54	28.88 +/- 0.95	0.00186641 +/- 0.00004697
success	3.4e-204	Betsimisaraka	Dai	Mozambique_Chopi	30.49	28.76 +/- 0.94	0.00196995 +/- 0.00005151
failure	1	Betsimisaraka	Esan	Yoruba	0.00	222.47 +/- inf	0.00000245 +/- inf
success	2e-10	Betsimisaraka	Esan	Mozambique_Chopi	6.36	97.58 +/- 15.34	0.00003802 +/- 0.00000588
success	5.2e-12	Betsimisaraka	Yoruba	Mozambique_Chopi	6.90	97.64 +/- 14.15	0.00003787 +/- 0.00000528
success	3.4e-06	Diego	Borneo	Sumba	4.64	39.49 +/- 8.50	0.00001391 +/- 0.00000250
success	0.045	Diego	Borneo	Dai	2.01	148.31 +/- 62.92	0.00000848 +/- 0.00000423
success	1.5e-144	Diego	Borneo	Esan	25.60	20.26 +/- 0.65	0.00146813 +/- 0.00005735
success	4.1e-148	Diego	Borneo	Yoruba	25.92	20.17 +/- 0.64	0.00145881 +/- 0.00005628
success	4.4e-159	Diego	Borneo	Mozambique_Chopi	26.87	20.26 +/- 0.64	0.00153534 +/- 0.00005713
success	0.0013	Diego	Sumba	Dai	3.21	54.62 +/- 16.99	0.00001871 +/- 0.00000415
success	2.2e-116	Diego	Sumba	Esan	22.93	20.57 +/- 0.73	0.00131657 +/- 0.00005741
success	9.8e-120	Diego	Sumba	Yoruba	23.27	20.46 +/- 0.71	0.00130741 +/- 0.00005619
success	6.7e-124	Diego	Sumba	Mozambique_Chopi	23.67	20.53 +/- 0.72	0.00137919 +/- 0.00005826
success	4.7e-130	Diego	Dai	Esan	24.26	20.24 +/- 0.66	0.00146933 +/- 0.00006055
success	1.1e-133	Diego	Dai	Yoruba	24.60	20.14 +/- 0.64	0.00145999 +/- 0.00005934
success	2.3e-138	Diego	Dai	Mozambique_Chopi	25.04	20.23 +/- 0.64	0.00153750 +/- 0.00006140
failure	0.073	Diego	Esan	Yoruba	1.80	187.29 +/- 74.66	0.00000278 +/- 0.00000155
success	2.8e-10	Diego	Esan	Mozambique_Chopi	6.31	138.53 +/- 20.05	0.00005992 +/- 0.00000950
success	6e-09	Diego	Yoruba	Mozambique_Chopi	5.82	141.26 +/- 20.77	0.00006105 +/- 0.00001049
success	3.1e-12	Merina	Borneo	Sumba	6.97	41.93 +/- 6.01	0.00002087 +/- 0.00000280
success	0.025	Merina	Borneo	Dai	2.24	130.22 +/- 46.24	0.00001713 +/- 0.00000766
success	1.2e-228	Merina	Borneo	Esan	32.28	22.65 +/- 0.70	0.00176093 +/- 0.00005003
success	5.5e-236	Merina	Borneo	Yoruba	32.80	22.66 +/- 0.69	0.00175499 +/- 0.00004855
success	1.5e-226	Merina	Borneo	Mozambique_Chopi	32.13	23.18 +/- 0.72	0.00186187 +/- 0.00004419
success	2.7e-09	Merina	Sumba	Dai	5.95	49.86 +/- 8.38	0.00002461 +/- 0.00000409
success	1.1e-227	Merina	Sumba	Esan	32.21	22.68 +/- 0.65	0.00154560 +/- 0.00004798
success	2.2e-238	Merina	Sumba	Yoruba	32.97	22.68 +/- 0.64	0.00153947 +/- 0.00004669
success	1.6e-250	Merina	Sumba	Mozambique_Chopi	33.81	23.32 +/- 0.69	0.00164251 +/- 0.00004452

Table 1: Supplementary Table 4: MALDER results for all Malagasy pop
lations, with varied African and Asian source populations. Columns are
follows: ... *(continued)*

test_res	p_value	test_pop	ref A	ref B	2-ref z-score	2-ref decay	2-ref amp_exp
success	3.8e-233	Merina	Dai	Esan	32.60	22.74 +/- 0.70	0.00176475 +/- 0.00005140
success	3.9e-244	Merina	Dai	Yoruba	33.37	22.75 +/- 0.68	0.00175932 +/- 0.00004984
success	1.8e-241	Merina	Dai	Mozambique_Chopi	33.19	23.29 +/- 0.70	0.00186752 +/- 0.00004541
failure	1	Merina	Esan	Yoruba	0.00	500.00 +/- inf	0.00001163 +/- inf
success	0.00052	Merina	Esan	Mozambique_Chopi	3.47	90.53 +/- 26.10	0.00002234 +/- 0.00000535
success	0.00013	Merina	Yoruba	Mozambique_Chopi	3.83	87.65 +/- 22.90	0.00002205 +/- 0.00000528
success	4.3e-06	MerinaA	Borneo	Sumba	4.59	41.16 +/- 8.96	0.00002155 +/- 0.00000380
success	0.029	MerinaA	Borneo	Dai	2.19	159.63 +/- 54.73	0.00002268 +/- 0.00001037
success	5.5e-141	MerinaA	Borneo	Esan	25.28	21.70 +/- 0.86	0.00172501 +/- 0.00006674
success	2.5e-139	MerinaA	Borneo	Yoruba	25.13	21.69 +/- 0.86	0.00172051 +/- 0.00006620
success	1.8e-135	MerinaA	Borneo	Mozambique_Chopi	24.77	21.86 +/- 0.88	0.00180736 +/- 0.00006513
success	3.3e-08	MerinaA	Sumba	Dai	5.53	50.19 +/- 9.08	0.00002540 +/- 0.00000437
success	1.8e-128	MerinaA	Sumba	Esan	24.11	21.96 +/- 0.86	0.00151669 +/- 0.00006290
success	4.2e-129	MerinaA	Sumba	Yoruba	24.17	21.94 +/- 0.86	0.00151185 +/- 0.00006254
success	8.1e-137	MerinaA	Sumba	Mozambique_Chopi	24.90	22.17 +/- 0.89	0.00159416 +/- 0.00006327
success	1.4e-149	MerinaA	Dai	Esan	26.05	21.80 +/- 0.84	0.00173212 +/- 0.00006550
success	1.3e-149	MerinaA	Dai	Yoruba	26.05	21.80 +/- 0.84	0.00172795 +/- 0.00006479
success	1.2e-146	MerinaA	Dai	Mozambique_Chopi	25.79	21.98 +/- 0.85	0.00181659 +/- 0.00006355
failure	1	MerinaA	Esan	Yoruba	0.00	244.30 +/- inf	0.00000233 +/- inf
success	0.021	MerinaA	Esan	Mozambique_Chopi	2.32	66.07 +/- 28.53	0.00001332 +/- 0.00000397
success	0.012	MerinaA	Yoruba	Mozambique_Chopi	2.51	57.29 +/- 22.87	0.00001150 +/- 0.00000378
success	3.3e-11	MerinaB	Borneo	Sumba	6.63	40.57 +/- 6.12	0.00001782 +/- 0.00000260
failure	0.072	MerinaB	Borneo	Dai	1.80	89.66 +/- 44.03	0.00000930 +/- 0.00000517
success	2.3e-53	MerinaB	Borneo	Esan	15.38	26.42 +/- 1.72	0.00182359 +/- 0.00006434
success	1.2e-58	MerinaB	Borneo	Yoruba	16.15	26.46 +/- 1.64	0.00181557 +/- 0.00006173
success	4.2e-61	MerinaB	Borneo	Mozambique_Chopi	16.49	26.93 +/- 1.63	0.00193439 +/- 0.00005846
success	0.034	MerinaB	Sumba	Dai	2.12	51.45 +/- 24.31	0.00002058 +/- 0.00000783
success	5.1e-52	MerinaB	Sumba	Esan	15.18	26.11 +/- 1.72	0.00159222 +/- 0.00006515
success	8.6e-57	MerinaB	Sumba	Yoruba	15.88	26.13 +/- 1.65	0.00158453 +/- 0.00006284
success	8.8e-60	MerinaB	Sumba	Mozambique_Chopi	16.31	26.75 +/- 1.64	0.00170122 +/- 0.00005956
success	1.9e-49	MerinaB	Dai	Esan	14.78	26.44 +/- 1.79	0.00181330 +/- 0.00006328
success	4.9e-54	MerinaB	Dai	Yoruba	15.48	26.48 +/- 1.71	0.00180611 +/- 0.00006077
success	3.8e-57	MerinaB	Dai	Mozambique_Chopi	15.93	26.94 +/- 1.69	0.00192369 +/- 0.00005553
failure	1	MerinaB	Esan	Yoruba	0.00	500.00 +/- inf	0.00001295 +/- inf
success	2.8e-06	MerinaB	Esan	Mozambique_Chopi	4.68	103.85 +/- 18.39	0.00003445 +/- 0.00000736
success	1.4e-06	MerinaB	Yoruba	Mozambique_Chopi	4.83	105.51 +/- 19.11	0.00003878 +/- 0.00000803
success	3.8e-08	Mikea	Borneo	Sumba	5.50	31.28 +/- 5.69	0.00001446 +/- 0.00000178
failure	0.25	Mikea	Borneo	Dai	1.14	218.09 +/- 127.53	0.00001453 +/- 0.00001274
success	6.2e-63	Mikea	Borneo	Esan	16.74	24.43 +/- 1.46	0.00179737 +/- 0.00007812
success	4.2e-61	Mikea	Borneo	Yoruba	16.49	24.45 +/- 1.48	0.00179523 +/- 0.00007745
success	8.6e-54	Mikea	Borneo	Mozambique_Chopi	15.44	23.99 +/- 1.55	0.00184531 +/- 0.00007972
success	9.4e-07	Mikea	Sumba	Dai	4.90	58.63 +/- 11.95	0.00002434 +/- 0.00000457
success	6.7e-59	Mikea	Sumba	Esan	16.18	24.75 +/- 1.53	0.00158879 +/- 0.00007863
success	1.3e-56	Mikea	Sumba	Yoruba	15.86	24.77 +/- 1.56	0.00158630 +/- 0.00007822
success	6.3e-50	Mikea	Sumba	Mozambique_Chopi	14.86	24.30 +/- 1.64	0.00163606 +/- 0.00008013
success	2.2e-58	Mikea	Dai	Esan	16.11	24.68 +/- 1.53	0.00181233 +/- 0.00008078
success	1.1e-56	Mikea	Dai	Yoruba	15.87	24.70 +/- 1.56	0.00181012 +/- 0.00008012
success	2.4e-50	Mikea	Dai	Mozambique_Chopi	14.92	24.24 +/- 1.62	0.00186109 +/- 0.00008254
failure	1	Mikea	Esan	Yoruba	0.00	186.33 +/- inf	0.00000210 +/- inf
success	0.00012	Mikea	Esan	Mozambique_Chopi	3.84	127.79 +/- 26.09	0.00004840 +/- 0.00001260
success	6.4e-07	Mikea	Yoruba	Mozambique_Chopi	4.98	113.09 +/- 19.86	0.00004262 +/- 0.00000856
success	4.8e-08	Sakalava_Tsimihety	Borneo	Sumba	5.46	36.93 +/- 6.76	0.00001530 +/- 0.00000123
success	0.023	Sakalava_Tsimihety	Borneo	Dai	2.27	143.03 +/- 63.01	0.00001010 +/- 0.00000364
success	1.1e-70	Sakalava_Tsimihety	Borneo	Esan	17.77	22.42 +/- 1.26	0.00177727 +/- 0.00007455

Table 1: Supplementary Table 4: MALDER results for all Malagasy pop
lations, with varied African and Asian source populations. Columns are
follows: ... *(continued)*

test_res	p_value	test_pop	ref A	ref B	2-ref z-score	2-ref decay	2-ref amp_exp
success	4.3e-69	Sakalava_Tsimihety	Borneo	Yoruba	17.57	22.33 +/- 1.27	0.00176631 +/- 0.00007391
success	3.3e-70	Sakalava_Tsimihety	Borneo	Mozambique_Chopi	17.71	23.09 +/- 1.30	0.00188537 +/- 0.00007484
success	0.0027	Sakalava_Tsimihety	Sumba	Dai	3.00	40.86 +/- 13.64	0.00001695 +/- 0.00000349
success	1.4e-73	Sakalava_Tsimihety	Sumba	Esan	18.14	22.70 +/- 1.25	0.00157199 +/- 0.00007657
success	3.5e-71	Sakalava_Tsimihety	Sumba	Yoruba	17.84	22.60 +/- 1.27	0.00156087 +/- 0.00007598
success	1.2e-73	Sakalava_Tsimihety	Sumba	Mozambique_Chopi	18.15	23.48 +/- 1.29	0.00167853 +/- 0.00007755
success	1.4e-72	Sakalava_Tsimihety	Dai	Esan	18.02	22.60 +/- 1.25	0.00177662 +/- 0.00007776
success	3.9e-71	Sakalava_Tsimihety	Dai	Yoruba	17.83	22.51 +/- 1.26	0.00176524 +/- 0.00007692
success	5.3e-72	Sakalava_Tsimihety	Dai	Mozambique_Chopi	17.94	23.26 +/- 1.30	0.00188434 +/- 0.00007879
failure	1	Sakalava_Tsimihety	Esan	Yoruba	0.00	500.00 +/- inf	0.00001659 +/- inf
success	7.2e-12	Sakalava_Tsimihety	Esan	Mozambique_Chopi	6.85	110.84 +/- 15.82	0.00004741 +/- 0.00000692
success	8.7e-14	Sakalava_Tsimihety	Yoruba	Mozambique_Chopi	7.46	109.22 +/- 14.49	0.00004564 +/- 0.00000612
success	5.9e-09	Temoro	Borneo	Sumba	5.82	45.13 +/- 7.76	0.00001730 +/- 0.00000224
failure	0.15	Temoro	Borneo	Dai	1.44	95.49 +/- 65.71	0.00000490 +/- 0.00000339
success	9.6e-133	Temoro	Borneo	Esan	24.52	29.11 +/- 0.86	0.00194337 +/- 0.00007926
success	1e-128	Temoro	Borneo	Yoruba	24.14	29.07 +/- 0.90	0.00193347 +/- 0.00008011
success	7.4e-147	Temoro	Borneo	Mozambique_Chopi	25.81	29.35 +/- 0.87	0.00205756 +/- 0.00007973
success	5e-07	Temoro	Sumba	Dai	5.03	54.64 +/- 10.87	0.00002150 +/- 0.00000353
success	5.3e-104	Temoro	Sumba	Esan	21.66	29.34 +/- 0.87	0.00172607 +/- 0.00007970
success	2.7e-101	Temoro	Sumba	Yoruba	21.37	29.28 +/- 0.90	0.00171615 +/- 0.00008032
success	7.5e-112	Temoro	Sumba	Mozambique_Chopi	22.47	29.60 +/- 0.86	0.00183343 +/- 0.00008158
success	5.4e-130	Temoro	Dai	Esan	24.26	29.48 +/- 0.82	0.00195981 +/- 0.00008079
success	6.1e-127	Temoro	Dai	Yoruba	23.97	29.42 +/- 0.85	0.00194883 +/- 0.00008131
success	2.8e-142	Temoro	Dai	Mozambique_Chopi	25.40	29.71 +/- 0.81	0.00207391 +/- 0.00008166
failure	1	Temoro	Esan	Yoruba	0.00	500.00 +/- inf	0.00000915 +/- inf
success	3.1e-05	Temoro	Esan	Mozambique_Chopi	4.17	139.01 +/- 29.38	0.00005303 +/- 0.00001272
success	1.2e-05	Temoro	Yoruba	Mozambique_Chopi	4.37	131.23 +/- 27.11	0.00004921 +/- 0.00001125
success	1.2e-07	Vezo	Borneo	Sumba	5.30	37.52 +/- 7.08	0.00001804 +/- 0.00000205
success	0.017	Vezo	Borneo	Dai	2.38	119.13 +/- 49.99	0.00000690 +/- 0.00000263
success	1.9e-69	Vezo	Borneo	Esan	17.62	21.98 +/- 1.25	0.00184944 +/- 0.00006413
success	5.5e-72	Vezo	Borneo	Yoruba	17.94	22.08 +/- 1.23	0.00184557 +/- 0.00006245
success	1.2e-72	Vezo	Borneo	Mozambique_Chopi	18.03	22.26 +/- 1.24	0.00194431 +/- 0.00005711
success	1.2e-08	Vezo	Sumba	Dai	5.70	44.79 +/- 7.86	0.00002035 +/- 0.00000314
success	2.4e-67	Vezo	Sumba	Esan	17.34	22.16 +/- 1.28	0.00163475 +/- 0.00005995
success	1.6e-70	Vezo	Sumba	Yoruba	17.76	22.26 +/- 1.25	0.00163029 +/- 0.00005825
success	1.3e-70	Vezo	Sumba	Mozambique_Chopi	17.76	22.41 +/- 1.26	0.00172420 +/- 0.00005415
success	4.2e-66	Vezo	Dai	Esan	17.17	22.04 +/- 1.28	0.00185290 +/- 0.00006451
success	1.1e-68	Vezo	Dai	Yoruba	17.51	22.13 +/- 1.26	0.00184873 +/- 0.00006279
success	3.5e-70	Vezo	Dai	Mozambique_Chopi	17.71	22.29 +/- 1.26	0.00194813 +/- 0.00005747
failure	0.13	Vezo	Esan	Yoruba	1.51	328.73 +/- 97.48	0.00000469 +/- 0.00000312
success	2.8e-05	Vezo	Esan	Mozambique_Chopi	4.19	117.74 +/- 23.59	0.00004669 +/- 0.00001114
success	2e-05	Vezo	Yoruba	Mozambique_Chopi	4.26	111.04 +/- 23.26	0.00004321 +/- 0.00001014

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